

Customer Data Quality Assessment for Pricing Intelligence

Online Retail II as Simulated Customer Data Handover

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Supporting notebook: <https://github.com/SkyFly03/Repeatable-Data-Quality-Framework-for-Pricing-Intelligence>

1. Goal and Approach

- Treat Online Retail II as a simulated first customer data handover.
- Assess whether the dataset is ready for pricing intelligence and model preparation.
- Identify issues that create revenue, cost-to-clean, customer service and model-readiness risks.
- Propose a repeatable intake specification and automated quality framework for future onboarding.

2. Key Data Quality Findings

Data Quality Issue	Finding	Business Impact	Priority
Missing Customer ID	22.77% of rows missing Customer ID	Blocks customer-level pricing, segmentation and customer behavior analysis	High
Missing product description	0.41% of rows missing Description	Weakens product validation and product-level analysis	Medium
Cancelled invoices	1.83% of rows	Must be separated from normal sales to avoid distorted revenue and demand signals	High
Negative quantities	2.15% of rows	Indicates returns/cancellations or invalid sales records	High
Zero or negative prices	0.58% of rows	Creates unreliable pricing and revenue calculations	Medium
Duplicate rows	3.22% of rows	Can overstate demand, revenue and customer activity	Medium
Product inconsistencies	1,232 StockCodes have multiple descriptions	Indicates weak product master data and risk for wrong product-level pricing	Medium
Quantity and price outliers	Approx. 1% high/low tails	May distort pricing patterns if not validated	Medium
Cost, margin, discount and contract fields	Not available	Blocks profitability-based pricing and price leakage analysis	Critical

3. Pricing Model Blockers and Commercial Risk

- Revenue risk: missing Customer ID, product hierarchy, cost, margin and discount context limit customer-level pricing, segmentation and margin improvement.
- Cost-to-clean risk: duplicates, inconsistent product descriptions and unclear transaction status create repeated manual cleaning work.
- Customer service risk: missing or unclear fields create clarification loops with the customer and slow onboarding.
- Model-readiness risk: cancellations, returns, duplicates, invalid values and outliers can distort demand, revenue and pricing signals.
- Margin leakage risk: revenue can be calculated, but profitability cannot be assessed without cost, margin, discount and contract field.

4. Proposed Customer Intake Specification

Required or Recommended Field

Required Field	Purpose
Customer_ID	Customer-level pricing and segmentation
Customer_Segment	Segment-based pricing logic
SKU_Product_ID	Product-level pricing and matching

Product_Description	Product validation
Product_Category	Product hierarchy and pricing groups
Invoice_ID and Invoice_Date	Transaction traceability and time-based pricing
Quantity and Unit_Price	Demand, volume and pricing analysis
Currency	Cross-market price comparison
Cost and Margin	Profitability-based pricing and margin protection
Discount and Contract_Price	Price leakage and contract pricing validation
Return_Cancellation_Flag	Separate normal sales from returns/cancellations
Country_Region and Sales_Channel	Market and channel-specific pricing
Source_System and Data_Extraction_Date	Repeatable onboarding and audit trail

5. Automated Checks to Run During Onboarding

- Completeness checks for all required fields.
- Duplicate row and duplicate invoice-line detection.
- Customer ID and SKU validation.
- Product code and description consistency check.
- Negative or zero quantity check.
- Zero, negative or extreme unit price check.
- Cancellation and return detection.
- Cost, margin and discount availability check.
- Quantity and price outlier checks.
- Source-system and extraction-date validation.
- Commercial risk classification by revenue readiness, cost-to-clean risk, customer service risk and model readiness.

6. Recommended First Improvement

Automated Customer Data Quality Scorecard Before Onboarding Starts

- Run the same quality checks for every new customer handover.
- Show whether the dataset is ready for pricing intelligence.
- Classify issues by revenue readiness, model readiness, cost-to-clean risk and customer service risk.
- Generate a clear customer feedback list before manual cleaning or model preparation begins.
- Reduce repeated manual work and make onboarding more scalable.

Expected business impact

- Faster customer onboarding.
- Lower manual cleaning effort.
- Better customer communication.
- More reliable pricing recommendations.
- Stronger foundation for margin improvement.

7. Example Customer Feedback

- Customer IDs are missing in **22.77%** of rows. Stable customer identifiers are required for customer-level pricing and segmentation.
- Cost, margin and discount fields are missing. These are required to identify margin leakage and pricing improvement potential.
- Cancellations and returns should be provided as a separate transaction status field, not mixed with normal sales.
- Product master data should be cleaned so each SKU has one stable product description and ideally one product category.
- Duplicate rows should be reviewed before pricing analysis because they can overstate demand, revenue and customer activity.

8. Method Note

- Analysis prepared with a reusable Python/pandas notebook in Google Colab, available via GitHub link.
- Notebook loads both yearly Excel sheets and applies completeness, consistency, duplicate, cancellation and outlier checks.
- Results are summarized into a repeatable onboarding scorecard.